

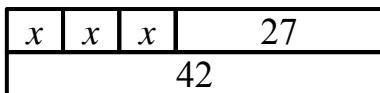


Equality in an equation

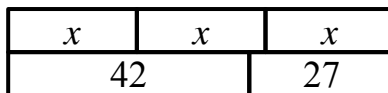
Task A: Using bar models to rearrange equations

1) Match the equations to the bar models. Each bar model can match with multiple equations.

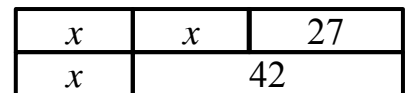
(A)



(B)



(C)



(D) $3x - 42 = 27$

(E) $3x = 42 + 27$

(F) $x = 42 - 27 - 2x$

(G) $27 = x + 42 - 2x$

(H) $2x + 27 - x = 42$

(I) $42 - 3x = 27$

(J) $27 + 3x = 42$

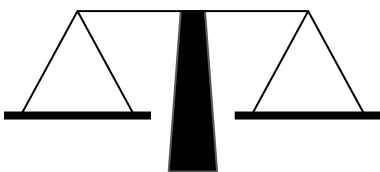
(K) $2x + x = 42 + 27$

2) For each bar model write another equation which is also true.

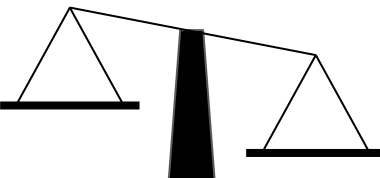
Task B: Using balances to represent equations

1) Choose one expression from the box to place on each scale pan.

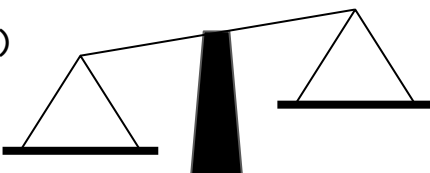
a)



b)



c)



$$8 + 10 + 13$$

$$a + 6 \qquad a + b$$

$$6 \times 5 \qquad 2a$$

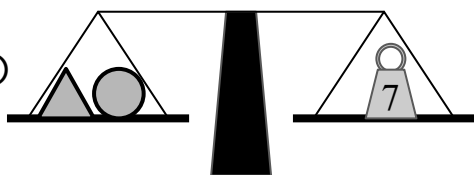
$$8 + 10 + 12$$

$$a \qquad 3 \times 5 \times 2$$

Explain your reasoning.

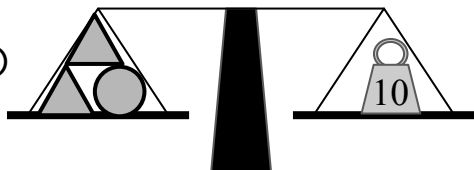
2)

a)



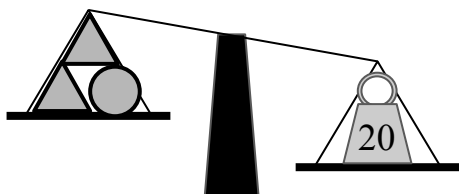
These scales are balanced. What might the mass of the triangle and circle be? Is there more than one answer?

b)



The scales are still balanced. What might the mass of the triangle and circle be? Is there more than one answer now?

c)



The scales are no longer balanced. What could be done to balance them? Is there more than one answer?

